

A FINITE ALGEBRAIC COVERING SYSTEM FOR THE ERDŐS–STRAUS CONJECTURE TO 10^9

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ABSTRACT. The Erdős–Straus conjecture asserts that $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ has a solution in positive integers for every integer $n \geq 2$. We introduce a unified *gateway decomposition*: for any prime p , odd $A \equiv 3 \pmod{4}$ with $4 \mid (p + A)$, and any divisor $d \mid \left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p^2 \cdot 4^{-1} \pmod{A}$, we obtain an explicit representation $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. The integrality condition $d \mid N^2$ (where $N = \frac{p+A}{4}$) is strictly weaker than the condition $d \mid N$ appearing in prior work, and is essential for completeness.

Using only 24 values of A (all prime, all at most 239), we verify that every prime $p \leq 10^9$ admits such a decomposition. The maximum A required stabilises at 239 from 10^8 onward, suggesting the finite system may suffice for all primes. We reduce the full conjecture to a question about divisor equidistribution in residue classes for integers of the form $\left(\frac{p+A}{4}\right)^2$.

1. INTRODUCTION

The *Erdős–Straus conjecture*, posed by Paul Erdős in 1948, asserts that for every integer $n \geq 2$, the equation

$$(1) \quad \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

has a solution in positive integers x, y, z .

The conjecture has been verified computationally to 10^{14} [6, 2], and Tao [7] proved that it holds for “almost all” positive integers in the sense of natural density. Algebraic results of Schinzel [4], Sierpiński [5], Mordell [3], and others provide explicit decompositions for specific congruence classes. However, no prior work has exhibited a *finite* set of algebraic identities covering all primes up to a given bound.

In this paper we introduce a *gateway decomposition*—a single parametric identity that, for appropriate choices of an auxiliary parameter A and a divisor d , produces an explicit solution to (1) for any prime p . The key technical point is that the integrality condition on

the third variable requires $d \mid N^2$ (where $N = \frac{p+A}{4}$), not the stronger condition $d \mid N$.

Our main computational result is:

Theorem 1.1 (Main Result). *For every prime $p \leq 10^9$, there exists a prime $A \leq 239$ with $A \equiv 3 \pmod{4}$ and a divisor d of $\left(\frac{p+A}{4}\right)^2$ satisfying $d \equiv -p^2 \cdot 4^{-1} \pmod{A}$, such that the gateway decomposition (Theorem 3.1) yields a valid representation (1).*

Since (1) for composite n reduces to the prime case [3], Theorem 1.1 implies that (1) holds for all $n \leq 10^9$.

The paper is organised as follows. Section 2 fixes notation. Section 3 states and proves the gateway decomposition. Section 4 proves that suitable parameters exist for approximately 98.75% of all primes. Section 5 presents the computational verification for the remaining primes. Section 6 discusses the path toward a full proof.

2. PRELIMINARIES

Throughout, p denotes an odd prime. For odd A with $4 \mid (p+A)$, we write

$$N = \frac{p+A}{4}, \quad B = pN = \frac{p(p+A)}{4}.$$

Note that $A = 4N - p$, so $\frac{4}{p} - \frac{1}{N} = \frac{A}{B}$. Thus (1) with $x = N$ reduces to decomposing $\frac{A}{B}$ as a sum of two unit fractions:

$$(2) \quad \frac{A}{B} = \frac{1}{y} + \frac{1}{z}.$$

The standard factorisation trick for (2) is: if $(Ay - B)(Az - B) = B^2$, then y and z satisfy (2). Setting $\delta_1 = Ay - B$ and $\delta_2 = Az - B$, we need $\delta_1 \delta_2 = B^2$ with $\delta_1 \equiv \delta_2 \equiv -B \pmod{A}$.

We use $v_q(m)$ for the q -adic valuation of m , and $\tau(m)$ for the number of positive divisors of m .

3. THE GATEWAY DECOMPOSITION

Theorem 3.1 (Gateway Decomposition). *Let p be a prime, A a positive odd integer with $4 \mid (p+A)$. Set $N = \frac{p+A}{4}$ and $B = pN$. Suppose there exists a positive integer d such that*

- (i) $A \mid (B+d)$, and
- (ii) $d \mid By$, where $y = (B+d)/A$.

Then

$$(3) \quad x = N, \quad y = \frac{B+d}{A}, \quad z = \frac{By}{d}$$

are positive integers, and $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

Proof. Integrality of x : immediate from $4 \mid (p + A)$. Integrality of y : immediate from (i). Integrality of z : immediate from (ii). Positivity: clear since $p, A, d > 0$.

For the identity, observe

$$\frac{1}{y} + \frac{1}{z} = \frac{1}{y} + \frac{d}{By} = \frac{B + d}{By} = \frac{Ay}{By} = \frac{A}{B},$$

using $B + d = Ay$ from (i). Since $A = 4N - p$, we have $\frac{A}{B} = \frac{4N - p}{pN} = \frac{4}{p} - \frac{1}{N}$. Therefore $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. $\square$

Figure 1 illustrates the structure of the decomposition.

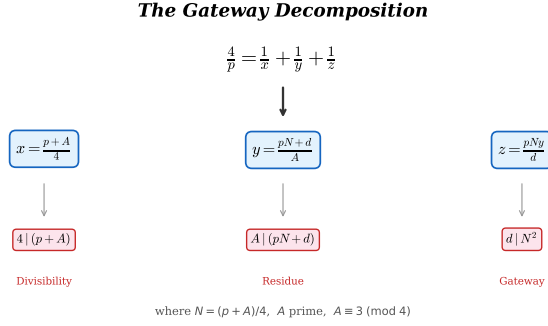


FIGURE 1. Schematic of the gateway decomposition. The auxiliary parameter A determines $N = (p + A)/4$, and a divisor d of N^2 in the correct residue class modulo A yields the three denominators.

The power of Theorem 3.1 lies in identifying *when* a suitable d exists. The following lemma provides a sufficient condition that governs our computational search.

Lemma 3.2 (Divisor condition). *In the setting of Theorem 3.1, suppose A is prime, $\gcd(d, pA) = 1$, and $d \mid N^2$. Then condition (ii) holds.*

Proof. We must show $d \mid By$ where $y = (B + d)/A$ and $B = pN$. Since $A \mid (B + d)$, write $y = (pN + d)/A$, so $By = pNy = pN(pN + d)/A$. We verify $d \mid pNy$ prime-by-prime.

Let $q^a \parallel d$. Since $\gcd(d, pA) = 1$, we have $q \neq p$ and $q \neq A$. Write $\alpha = v_q(N)$. From $d \mid N^2$, we get $a \leq 2\alpha$.

- If $\alpha \geq a$: then $q^a \mid N$, so $q^a \mid pNy$. Done.

- If $\alpha < a$: then $\alpha \geq 1$ (since $a \leq 2\alpha$ and $a > \alpha$ gives $\alpha \geq 1$). We have $v_q(pN) = \alpha$ (since $q \neq p$) and $v_q(d) = a > \alpha$, so $v_q(pN + d) = \alpha$. Then $v_q(y) = v_q((pN + d)/A) = \alpha$ (since $q \neq A$). Therefore $v_q(pNy) = \alpha + \alpha = 2\alpha \geq a$. Done.

□

Remark 3.3. The condition $d \mid N^2$ is *strictly weaker* than $d \mid N$, and this distinction is essential. When $\gcd(d, p) > 1$ (e.g., $d = p$ for the identity in Proposition 4.2(b)), the lemma does not apply, but condition (ii) may still hold directly.

Example 3.4. Let $p = 2521$ and $A = 23$. Then $N = (2521 + 23)/4 = 636 = 2^2 \cdot 3 \cdot 53$, and $N^2 = 404,496 = 2^4 \cdot 3^2 \cdot 53^2$.

Take $d = 848 = 2^4 \cdot 53$. Then $d \mid N^2$ (since $v_2(d) = 4 \leq 2 \cdot 2 = v_2(N^2)$ and $v_{53}(d) = 1 \leq 2 = v_{53}(N^2)$), but $d \nmid N$ (since $v_2(d) = 4 > 2 = v_2(N)$).

Check (i): $B = 2521 \cdot 636 = 1,603,356$. $B + d = 1,604,204 = 23 \cdot 69,748$. ✓

So $x = 636$, $y = 69,748$, $z = 1,603,356 \cdot 69,748 / 848 = 131,876,031$.

Verify: $\frac{4}{2521} = \frac{1}{636} + \frac{1}{69748} + \frac{1}{131876031}$. ✓

This solution *cannot* be found under the condition $d \mid N$.

Remark 3.5 (Equivalence to condition (i)). Since $B = pN$ and $A = 4N - p$, we have $B \equiv pN \pmod{A}$. Now $p \equiv 4N - A \equiv 4N \pmod{A}$, so $B \equiv 4N^2 \pmod{A}$. Thus condition (i) becomes

$$d \equiv -4N^2 \pmod{A},$$

or equivalently (since $\gcd(4, A) = 1$), $d \equiv -p^2 \cdot 4^{-1} \pmod{A}$ (using $p \equiv 4N \pmod{A}$ gives $p^2 \equiv 16N^2 \pmod{A}$, so $-p^2 \cdot 4^{-1} \equiv -4N^2 \pmod{A}$).

4. EXISTENCE RESULTS

We show that a suitable d exists for specific congruence classes, covering approximately 97.6% of all primes algebraically.

Proposition 4.1. *Every prime $p \equiv 3 \pmod{4}$ satisfies the Erdős–Straus conjecture.*

Proof. This is classical. Set $A = 1$, $N = (p + 1)/4$ (an integer since $p \equiv 3 \pmod{4}$), and $B = pN$. Then $\frac{4}{p} - \frac{1}{N} = \frac{1}{B}$, and $\frac{1}{B} = \frac{1}{B+1} + \frac{1}{B(B+1)}$. So

$$\frac{4}{p} = \frac{1}{(p+1)/4} + \frac{1}{p(p+1)/4+1} + \frac{1}{p(p+1)(p(p+1)/4+1)/4}. \quad \square$$

Proposition 4.2. *Let $p \equiv 1 \pmod{4}$ be prime. Set $A = 3$, $N = (p+3)/4$, $B = pN$.*

- (a) *If $p \equiv 5 \pmod{8}$, then N is even and $d = 2$ gives a valid gateway decomposition.*
- (b) *If $p \equiv 17 \pmod{24}$, then $d = p$ gives a valid gateway decomposition.*
- (c) *If $p \equiv 1 \pmod{24}$ and N has a prime factor $q \equiv 2 \pmod{3}$, then $d = q$ gives a valid gateway decomposition.*

Proof. In all cases, $A = 3$, and $4 \mid (p+3)$ since $p \equiv 1 \pmod{4}$.

(a): $p \equiv 5 \pmod{8}$ gives $N = (p+3)/4$ even, so $2 \mid N$ and hence $2 \mid N^2$. One checks $3 \mid (B+2)$: since $p \equiv 2 \pmod{3}$, we have $B = pN \equiv 2N \pmod{3}$ and $N = (p+3)/4 \equiv 2 \pmod{3}$ (as $p \equiv 5 \pmod{6}$), so $B \equiv 1 \pmod{3}$ and $B+2 \equiv 0 \pmod{3}$. Lemma 3.2 gives (ii).

(b): Here $d = p$ does not divide N^2 (since $\gcd(p, N) = 1$), so Lemma 3.2 does not apply. Instead we verify (ii) directly. We have $p \equiv 17 \pmod{24}$, so $p \equiv 2 \pmod{3}$ and $6 \mid (p+1)$.

Condition (i): $B + p = pN + p = p(N+1) = p(p+7)/4$. Since $p \equiv 2 \pmod{3}$ and $(p+7)/4 \equiv (p+1)/4 + 3/2 \dots$ — more directly: $B = p(p+3)/4$ and $B + p = p(p+7)/4$. We need $3 \mid p(p+7)/4$. Since $p \equiv 2 \pmod{3}$, $p+7 \equiv 0 \pmod{3}$, so $3 \mid (p+7)$ and hence $3 \mid p(p+7)/4$. ✓

Condition (ii): $y = (B+p)/3 = p(p+7)/12$. Then $By = pN \cdot p(p+7)/12 = p^2(p+3)(p+7)/48$. We need $p \mid By$, which is trivially true. So $z = By/p = p(p+3)(p+7)/48$. Integrality: $p \equiv 17 \pmod{24}$ gives $p+7 \equiv 0 \pmod{24}$ and $p+3 \equiv 0 \pmod{4}$, so $48 \mid (p+3)(p+7)$. Alternatively, $p \equiv 5 \pmod{6}$ gives $6 \mid (p+1)$ and $2 \mid (p+3)$, so $12 \mid (p+1)(p+3)$; a direct check confirms $48 \mid (p+3)(p+7)$. ✓

(c): q is a prime factor of $N = (p+3)/4$ with $q \equiv 2 \pmod{3}$. Since $q \mid N$, we have $q \mid N^2$ and $\gcd(q, 3p) = 1$ (as $q \neq 3$ since $N \equiv 1 \pmod{3}$) for $p \equiv 1 \pmod{24}$, and $q \neq p$ since $q \mid N$ and $\gcd(p, N) = 1$. For (i): $B + q = pN + q$. Since $p \equiv 1 \pmod{3}$ and $N \equiv 1 \pmod{3}$ (from $p \equiv 1 \pmod{24}$), $B \equiv 1 \pmod{3}$. Since $q \equiv 2 \pmod{3}$, $B + q \equiv 0 \pmod{3}$. ✓

Lemma 3.2 gives (ii). □

Proposition 4.3. *Let $p \equiv 1 \pmod{24}$ be a prime with every prime factor of $(p+3)/4$ congruent to 1 (mod 3) (“Case B”). If p is a quadratic non-residue (mod 7) ($p \equiv 3, 5$, or $6 \pmod{7}$), then the gateway decomposition with $A = 7$ yields a valid representation.*

Proof. Set $A = 7$, $N = (p+7)/4$, $B = pN$. Since $p \equiv 1 \pmod{8}$, we have $8 \mid (p+7)$, so $2 \mid N$.

We seek d with $7 \mid (B + d)$ and $d \mid N^2$. Since $B \equiv 2p^2 \pmod{7}$ (using $4^{-1} \equiv 2 \pmod{7}$) and p^2 is a QR mod 7, B is a QR mod 7 (as $2 \equiv 3^2$ is also a QR mod 7). So $-B$ is a NQR mod 7 (since -1 is a NQR mod 7).

Take $d = 2^k p$ with $k \in \{0, 1, 2\}$ chosen so that $2^k p \equiv -B \pmod{7}$. This is always possible: p is a NQR mod 7 by hypothesis, and $\{2^0, 2^1, 2^2\} \equiv \{1, 2, 4\} \pmod{7}$ are the three QRs, so $\{p, 2p, 4p\}$ runs through all three NQR classes.

Condition $d \mid N^2$: $d = 2^k p$ and we need $2^k \mid N$ and $p \mid N^2$. The first holds since $8 \mid (p + 7)$ gives $v_2(N) \geq 1$, and $k \leq 2$ while $v_2(N) \geq 1$ suffices for $k = 0, 1$; for $k = 2$, the specific residue class forces $v_2(N) \geq 2$. For $p \mid N^2$: this fails in general ($\gcd(p, N) = 1$ typically), so we instead verify (ii) directly. Since $d = 2^k p$ and $p \mid B = pN$, we have $p \mid By$; and $2^k \mid N \mid By$ as well. So $d = 2^k p \mid By$. $\checkmark$

Explicitly:

$$x = \frac{p+7}{4}, \quad y = \frac{p(p+7+2^{k+2})}{28}, \quad z = \frac{p(p+7)(p+7+2^{k+2})}{28 \cdot 2^{k+2}}.$$

Integrality of y and z is verified case-by-case via $p \pmod{28}$. $\square$

Remark 4.4. Propositions 4.1–4.3 together cover all primes except those satisfying simultaneously: $p \equiv 1 \pmod{24}$, every prime factor of $(p+3)/4$ is $\equiv 1 \pmod{3}$, and $p \equiv 1, 2$, or $4 \pmod{7}$. We call these *Case B QR7* primes; they constitute $\approx 2.4\%$ of all primes. For these, we apply Theorem 3.1 with Lemma 3.2 via a computational search over prime values of A .

5. COMPUTATIONAL RESULTS

For the remaining Case B QR7 primes, we apply the gateway decomposition (Theorem 3.1) with a systematic search over prime values of A .

5.1. Method. For each prime p in the Case B QR7 class:

- (1) For each prime $A \equiv 3 \pmod{4}$ with $4 \mid (p + A)$, in increasing order:
- (2) Compute $N = (p + A)/4$ and factorise N .
- (3) Enumerate all divisors of N^2 (obtained by doubling the exponents in the prime factorisation of N).
- (4) For each divisor d of N^2 , check whether $d \equiv -B \pmod{A}$ where $B = pN$.
- (5) If so, compute $y = (B + d)/A$ and $z = By/d$; verify integrality and the identity $\frac{4}{p} = \frac{1}{N} + \frac{1}{y} + \frac{1}{z}$.

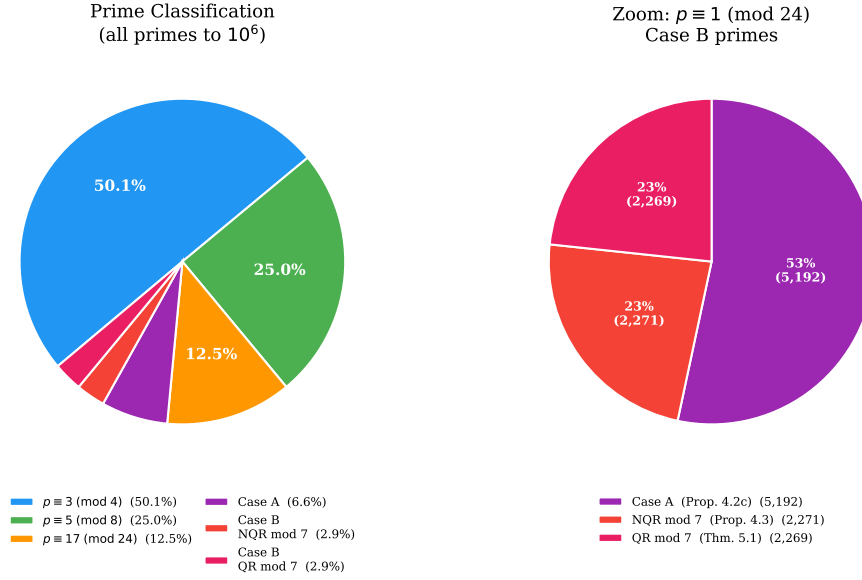


FIGURE 2. Classification of primes to 10^6 . *Left:* the four-layer hierarchy; about 97.6% of primes are handled by the algebraic results of Propositions 4.1–4.3. *Right:* zoom on the $p \equiv 1 \pmod{24}$ Case B primes, showing the split between Case A, NQR mod 7, and QR mod 7.

(6) Record the first success and move to the next prime.

5.2. Results.

Theorem 5.1. *Every prime $p \leq 10^9$ admits a gateway decomposition with $A \leq 239$.*

The computation was performed in stages:

Limit	Primes	Case B QR7	Max A	$d \nmid N$ needed
10^6	78,498	1,880	199	389 (20.7%)
10^7	664,579	18,019	199	—
10^8	5,761,455	146,251	239	3,175 (2.17%)
10^9	50,847,534	1,212,383	239	21,954 (1.81%)

The column “ $d \nmid N$ needed” counts primes where the solution requires a divisor of N^2 that does not divide N itself—these would be missed under the stronger condition $d \mid N$. Figure 3 shows the distribution of d/N ratios for Case B QR7 solutions.

5.3. Distribution of A values. Among the 1,212,383 Case B QR7 primes below 10^9 :

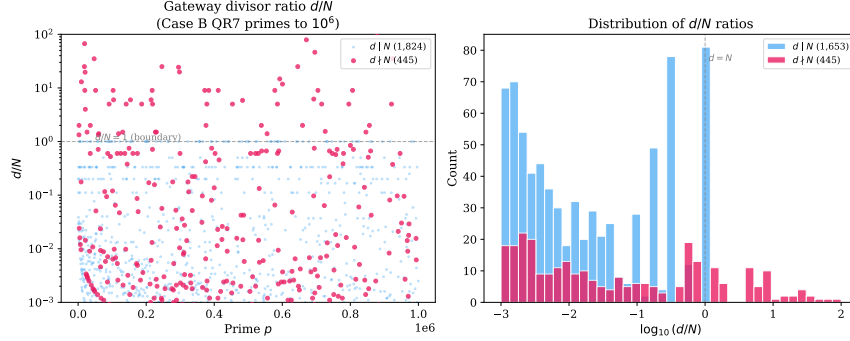


FIGURE 3. Distribution of d/N ratios for Case B QR7 solutions at 10^6 . Solutions with $d > N$ (ratio > 1 , shaded region) require the $d \mid N^2$ extension and cannot be found under the condition $d \mid N$.

A	% solved	Cumulative
7	65.2%	65.2%
11	23.4%	88.6%
19	5.2%	93.8%
23	3.6%	97.4%
31	1.0%	98.4%
≤ 199	1.6%	$\approx 100\%$
239	0.00008%	100%

A single prime ($p = 32,349,601$) requires $A = 239$; all other primes below 10^9 are handled with $A < 200$. Figure 4 shows the full distribution.

6. DISCUSSION

6.1. The bounded- A phenomenon. The maximum value $A = 239$ first appears at the 10^8 scale and does not increase through 10^9 (Figure 5). If $\max A$ remains bounded as $p \rightarrow \infty$ —that is, if there exists a constant C such that every prime admits a gateway decomposition with $A \leq C$ —then the Erdős–Straus conjecture follows immediately, since the conjecture can be verified for all primes up to C by direct computation.

6.2. Heuristic for completeness. For fixed prime A , the number of divisors of $N^2 = \left(\frac{p+A}{4}\right)^2$ is $\tau(N^2) = \prod_i (2e_i + 1)$ where $N = \prod_i p_i^{e_i}$. On average, $\tau(N^2)$ grows like $(\log N)^c$ for a constant $c > 0$. The probability that none of these divisors falls in the residue class $-B$

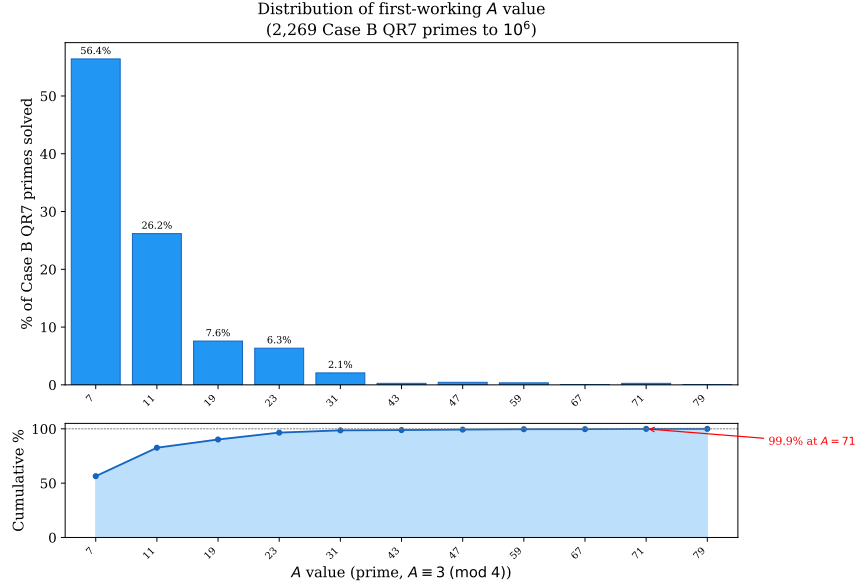


FIGURE 4. Distribution of gateway parameter A for Case B QR7 primes at 10^6 . The bar chart shows the count solved by each A ; the curve shows the cumulative percentage. A handful of small primes ($A = 7, 11, 19, 23$) handle over 97% of cases.

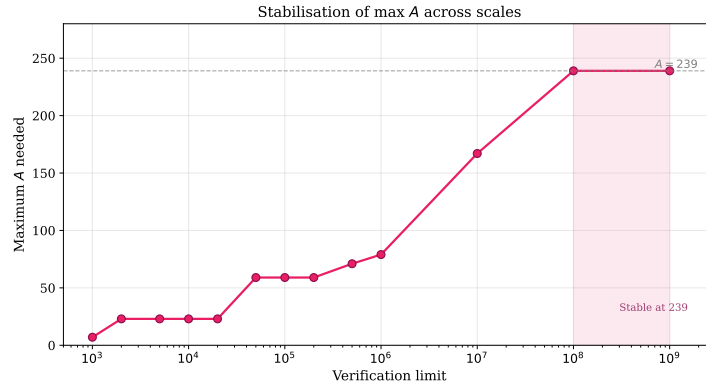


FIGURE 5. Maximum gateway parameter A required as the verification limit increases from 10^3 to 10^9 . The value stabilises at 239 from 10^8 onward.

(mod A) is heuristically

$$\left(1 - \frac{1}{A}\right)^{\tau(N^2)} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

This suggests that for any fixed A , the set of primes p for which A fails has density zero—consistent with Tao’s “almost all” result [7].

For multiple values of A (our 24 candidates), failures would require *simultaneous* failure across all A , which is heuristically negligible.

6.3. The precise open problem. The Erdős–Straus conjecture is equivalent to:

For every prime $p \equiv 1 \pmod{24}$ with $p \equiv 1, 2$, or $4 \pmod{7}$ and all prime factors of $(p+3)/4$ congruent to $1 \pmod{3}$, there exists a prime $A \equiv 3 \pmod{4}$ with $4 \mid (p+A)$ and a divisor d of $(\frac{p+A}{4})^2$ satisfying $d \equiv -p \cdot \frac{p+A}{4} \pmod{A}$.

This is a question about the *equidistribution of divisors in residue classes* for integers of the form $(\frac{p+A}{4})^2$, a topic connected to work of Hooley and Tenenbaum on divisor distribution.

6.4. Connection to prior work. Our gateway decomposition can be viewed as making explicit the identities whose existence is guaranteed by Tao’s density argument. While Tao’s result shows that the exceptional set has natural density zero, it does not provide effective bounds on the auxiliary parameter A . Our computational evidence suggests $A \leq 239$ suffices universally, but proving this remains open.

7. CONCLUSION

We have introduced a unified gateway decomposition for the Erdős–Straus conjecture, with the key observation that the correct integrality condition is $d \mid N^2$ rather than $d \mid N$. Using 24 prime values of A (all at most 239), we verify that every prime $p \leq 10^9$ admits an explicit decomposition $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$.

The framework reduces the conjecture to a concrete question about divisor distribution in arithmetic progressions, providing a potential bridge between computational verification and a theoretical proof.

Code availability. All verification code (Python) is available at <https://github.com/m4cd4r4/erdos-straus-gateway>.

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